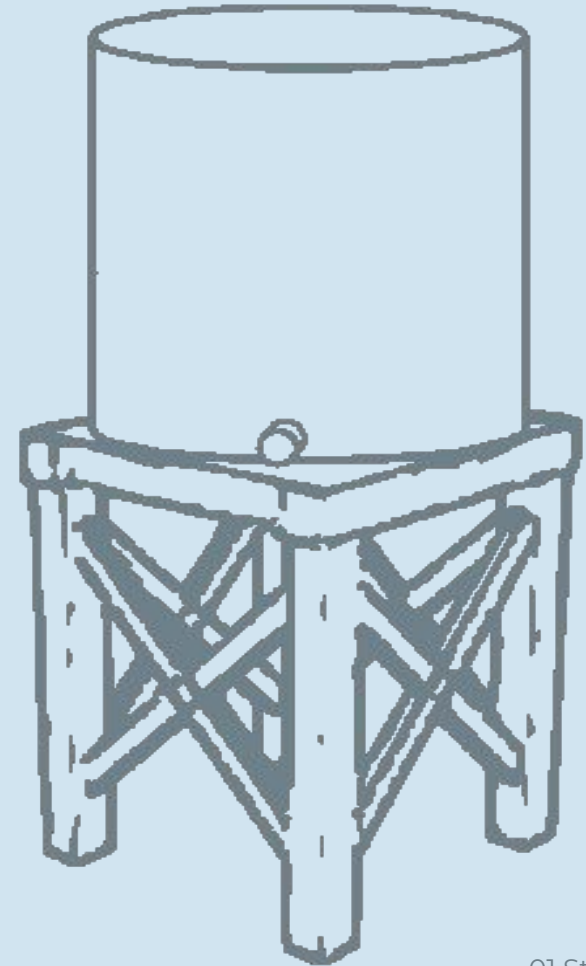


RAINWATER HARVESTER SYSTEM PROPOSAL

Water from sky to sip

Team A6

Ming Gao, Kyrin Chen, Marcus Kwong,
Adrian Lee, Charles Xiong, Si Wu





Coastal community of Van Anda, BC

INTRODUCTION

The Problem

Remote communities like Van Anda can face limited access to safe and reliable drinking water. Our system must provide an off-grid solution that captures, stores, treats, and delivers clean water effectively.

Project Objectives

- Supply potable water for a 2-person household
- Ensure reliable year-round off-grid operation
- Optimize cost, safety, maintenance, and sustainability

DESIGN PROCESS

02 Component Selection

Studied possible components to brainstorm and explore diverse solutions



01 Studying the Problem

Identified stakeholder needs for remote communities to establish design goals



Iteration

Explored alternative solutions and configurations

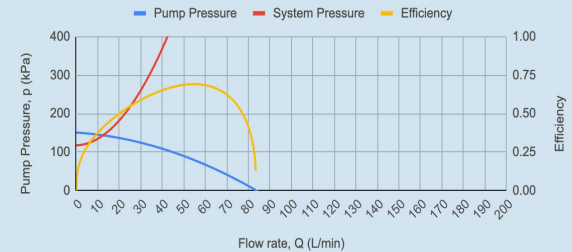
03 Modelling and Simulations

Modelled systems performance by using historical weather data

04 Risk and Sustainability

Quantified potential failures and calculated environmental impacts of long-term operation

Pump, System, and Efficiency Curve vs. Flow Rate (Q)



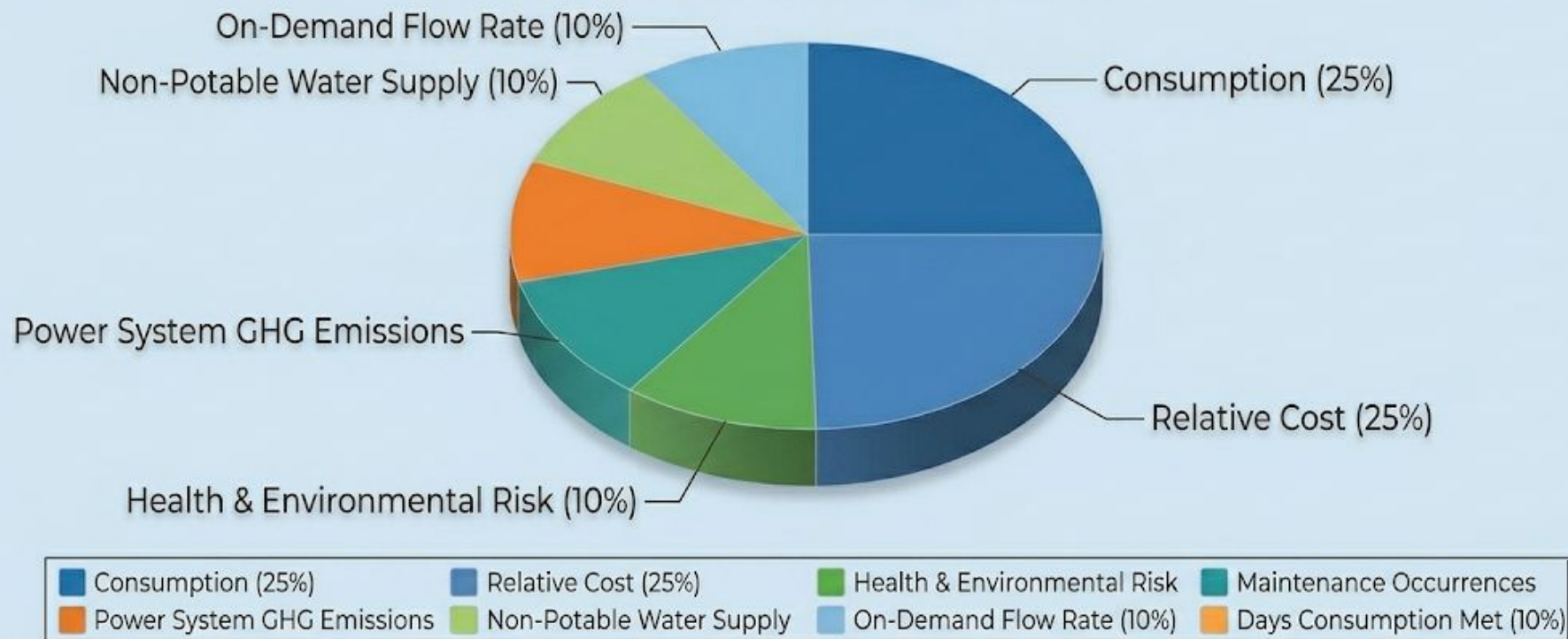
05 Final System Development

Optimized system design to ensure maximum stakeholder satisfaction

STAKEHOLDER CONSULTATION

Attributes	Min.	Max	Units	Weight
Consumption	125	745	L/day	25%
Relative Cost	30	115	% relative to shipping water	25%
Health & Environmental Risk	1	24	N/A	10%
Power System GHG Emissions	15	110	% relative to shipping water	10%
Maintenance Occurrences	10	60	#/year	10%
Non-Potable Water Supply	0	0.3	Fraction of Consumption	5%
On-Demand Flow Rate	18	40	L/min	5%
Days Consumption Met	200	365	Days/year	10%

Weight Contribution by Attribute



OPTIMAL DESIGN

Total Cost (5 years operation)
\$53,453.37
GHG Emissions (5 years)
4608[kgCO2e]
Satisfaction
72.89%

Consumption	600L
Non-Potable Water Use	0L
Roof Catchment	Whole Roof
Extra Catchment Area	80m ² (x: 25m, y: 25m, z: 2.5m)
Collection Tank Volume	5000L
Storage Tank Volume	39m ³ (x: 25m y: 0m z: 2.5m)H: 8m
Pump Selection	A
Filters	After storagewith 1, 5 & 200 micron
UV Disinfection	40W
Chemical Disinfection	Chlorine
Power System	Solar Panels (HES-305P), (3)
# of Batteries	1
Cf_exp	12052

PROTOTYPES & TESTING

Simulating Rainwater Storage & Estimating Catchment Tank Volume Satisfaction using Historical Weather Data.

Consumption values used:

~1% satisfaction:	166	L/day
~50% satisfaction:	369	L/day
~99% satisfaction:	572	L/day

V_{tank} values (in m³):

	~1% Satisfaction	~50% Satisfaction	~99% Satisfaction
Station 1 – 2014	5.86	16.62	39.86
Station 1 – 2015	7.52	28.67	52.01
Station 2 – 2015	8.34	Not possible	Not possible
Station 3 – 2013	4.96	Not possible	Not possible

C_{f,exp} Flow Lab Experiment

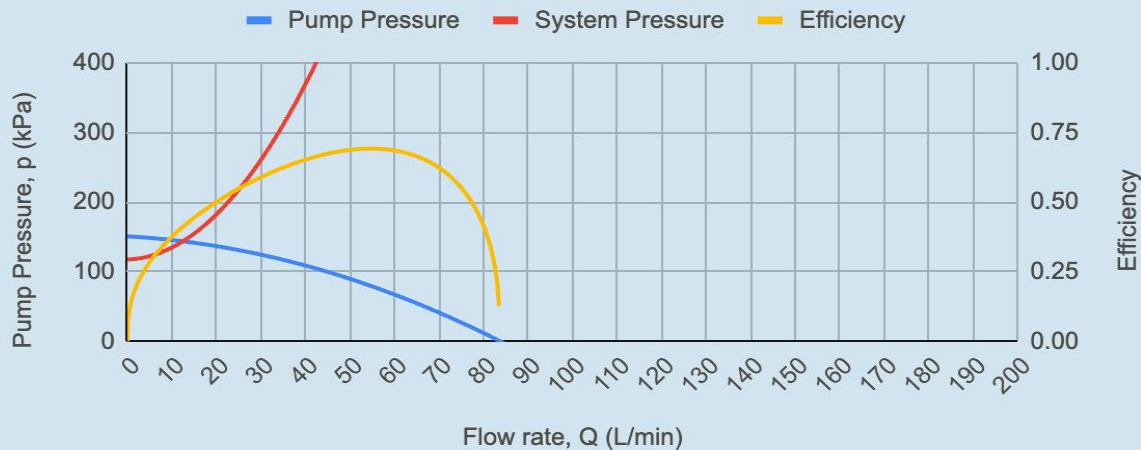


RWH System Model Parameter Testing

VARIABLES	VALUES	UNITS	SYMBOLS
a	-0.0172		
b	-0.3605		
c	150.9		
A	0.7		
B	0.39		
Maximum flow rate	83.7	L/min	Q _{max}
Length of piping to storage	40.31128874	m	L
Loss coefficient, catchment to storage	8		K
Total filter coefficient	3066.87244		C _f
h _{storage} going to storage	12	m	h _{storage, pump}
Gravity acceleration	9.81	m/s ²	g
Water density	1000	kg/m ³	ρ (rho)
Pipe diameter	20	mm	D
Pipe friction factor	0.05		f
Flow velocity step size	0.01	m/s	

PUMP SPECIFICATIONS

Pump, System, and Efficiency Curve vs. Flow Rate (Q)



Pump A

- Pump A was selected by comparing the pump curve with the system pressure curve
- The curve intersection shows the pump can overcome elevation and pipe losses
- Pump efficiency was also considered to reduce energy use
- Overall, Pump A provided the best balance of pressure, flow, and efficiency

Operating flow rate
15.83362697
Operating pressure
140.8798731
Operating efficiency
0.4536538167

FILTRATION / DISINFECTION

- 200 μm , 5 μm , and 1 μm filters provide staged particle removal
- Chlorine disinfects water before storage
- UV provides final treatment before water enters the house
- This setup improves water quality and system reliability



Bag Filter (200 μm)



Cartridge Filter (5 μm)



Cartridge Filter (1 μm)

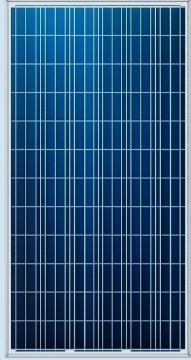


UV Lamp (40W)



Chlorine Disinfection System

POWER / ENERGY



HES-305P Solar Panel x 3

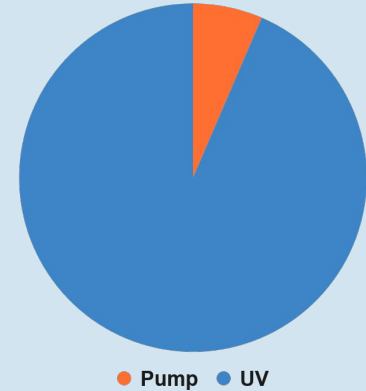
- Captures solar energy to power system operations and charge the battery bank.



12V Deep Cycle Wet Lead Acid Battery

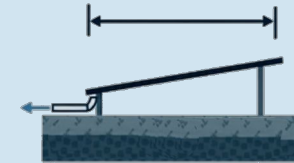
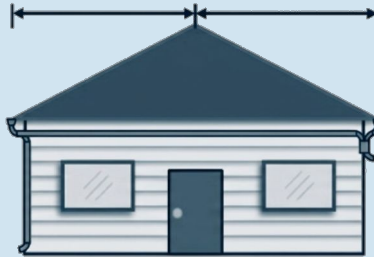
- Provides resilient energy storage solution to support peak load demands

Our system uses an average of 4.92MJ of energy daily.

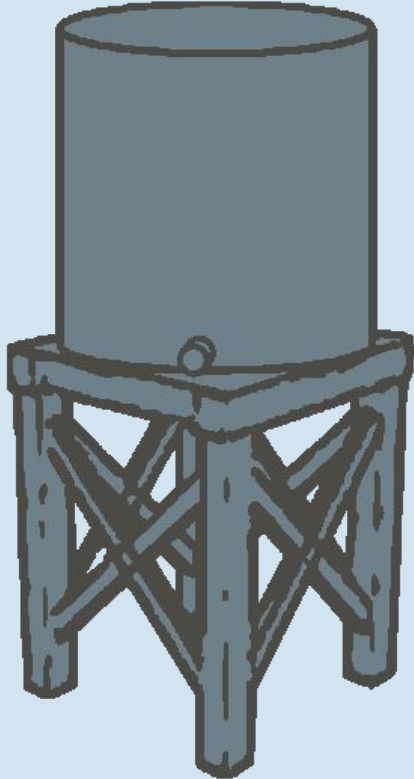


CATCHMENT & CONTAINMENT

- Rainwater is collected from the entire 100m² roof and an additional 80m² catchment surface
- This gives a total catchment area of 180m², increasing water collection during wet periods
- A 5000L catchment tank was selected as a practical balance between storage capacity, space, and cost



STORAGE TANK



- Rainwater is collected from the entire 100 m² roof and an additional 80 m² catchment surface
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- A 5000 L catchment tank was selected as a practical balance between storage capacity, space, and cost

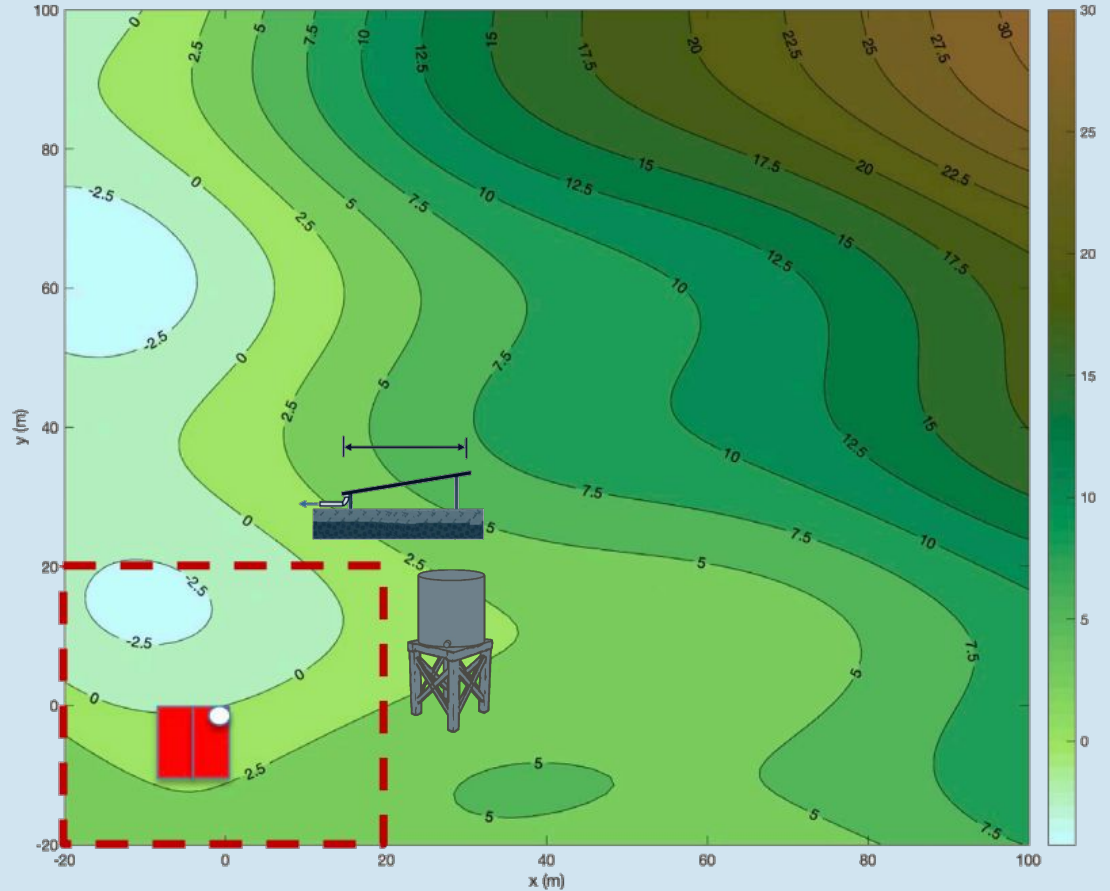
SYSTEM MAP

- Storage Tank

x	y	z
25m	0m	2.5m

- Catchment Area

x	y	z
25m	25m	2.5m



RISK ANALYSIS & MANAGEMENT

Main system risks were low-rainfall shortages, reduced flow from filter clogging, and power system downtime

Risk was managed through 180 m² catchment, 39 m³ storage, 200/5/1 µm staged filtration, and generator + battery support

Our final design emphasizes reliability, maintainability, and safe operation in a remote setting

Risk	Impact	Mitigation
Low rainfall / dry periods	Tank runs low, lower reliability	Larger storage tank + added catchment
Filter fouling	Lower flow, weaker treatment performance	200 µm + 5 µm + 1 µm staged filtration
Power interruption	Pump/UV cannot operate properly	Solar panels + 1 battery

GREENHOUSE GAS EMISSIONS

GHG Emissions (5 years)	Relative GHG Emissions (existing system: 9525.5 [kg CO ₂ e])
4608kg CO ₂ e	48.38%

Storage System
2408kg CO₂e

Solar Panels
1860kg CO₂e

Battery
240kg CO₂e

Power Inverter
100kg CO₂e

STAKEHOLDER SATISFACTION

Attributes	Min.	Max	Value	Satis.	Weight
Consumption	125	745	600	87%	25%
Relative Cost	30	115	48.59	89%	25%
Health & Environmental Risk	1	24	15.56	30%	10%
Power System GHG Emissions	15	110	48.38	73%	10%
Maintenance Occurrences	10	60	32.8	57%	10%
Non-Potable Water Supply	0	0.3	0	0%	5%
On-Demand Flow Rate	18	40	30.52	61%	5%
Days Consumption Met	200	365	365	100%	10%

- Calculated using established equations
- Input values from our simulations

Overall Satisfaction
72.89%

(1) ALTERNATE CONFIGURATION

Total Cost (w 5 years operation)
\$51,643.30
GHG Emissions (5 years)
5985.4[kgCO2e]
Satisfaction
63.39%

Consumption	600L
Non-Potable Water Use	0L
Roof Catchment	Whole Roof
Extra Catchment Area	86m ² (x: 20m y: 20m z: 2.5m)
Collection Tank Volume	5000L
Storage Tank Volume	33m ³ (x: 20m y: 0m z: 2.5m)
Pump Selection	A
Filters	1, 5 & 200 micron
UV Disinfection	36W
Chemical Disinfection	Chlorine
Power System	Diesel
# of Batteries	1
Cf_exp	12052

(2) ALTERNATE CONFIGURATION

Total Cost (/w 5 years operation)
\$43,824.40
GHG Emissions (5 years)
4608[kgCO2e]
Satisfaction
68.49%

Consumption	580L
Non-Potable Water Use	0L
Roof Catchment	Whole Roof
Extra Catchment Area	80m^2 (x: 20m y: 0m z: 2.5m)
Collection Tank Volume	5000L
Storage Tank Volume	33m³ (x: 20m y: 0m z: 2.5m)
Pump Selection	C
Filters	1 & 200 micron
UV Disinfection	36W
Chemical Disinfection	Chlorine
Power System	Solar Panels (HES-305P), (3)
# of Batteries	1
Cf_exp	12052

CONCLUSION

Set Up Cost	Cyclic Cost (5 years)	GHG Emissions (5 yrs)	Satisfaction
\$38878.37	\$14575.00	9525.5[kgCO ₂ e]	72.89%

Raw Water — 200µm Filter — 5µm — 1µm — Chlorine — UV — Tap.

This system is able to deliver a safe, affordable, and sustainable solution for every family in the community.

THANK YOU

Together, we can build a more resilient and sustainable future for our remote communities.

Team A6

Ming Gao, Kyrin Chen, Marcus Kwong, Adrian Lee, Charles Xiong, Si Wu

GenAI ACKNOWLEDGEMENT

Generative AI was used throughout the production of this presentation as a tool to assist and refine in the following ways:

- To **remove backgrounds**,
- To **recolour**, and **upscale images**
- To **create charts**
- To **proofread** for **grammar, conventions**, and **conversational flow**